OPEN SOURCE TECHNOLOGIES (01CE0618) Lab Manual

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Experiment 1

AIM: Explore GitHub/GitLab for open-source projects with different licenses.

1. List of GitHub Licenses:

- No License
- Apache License 2.0
- MIT License
- BSD 2-Clause License
- BSD 3-Clause License
- Boost Software License 1.0
- Creative Commons Zero (CC0) 1.0
- Eclipse Public License 2.0
- GNU General Public License v2.0 (GPL-2.0)
- GNU General Public License v3.0 (GPL-3.0)
- GNU Lesser General Public License v2.1 (LGPL-2.1)
- GNU Affero General Public License v3.0 (AGPL-3.0)
- Mozilla Public License 2.0 (MPL-2.0)
- The Unlicense

2. Licenses and Short Description Table:

License	Short Description
No License	Code is fully copyrighted; others cannot legally use, modify, or distribute it.
Apache License 2.0	Permissive license allowing commercial use, modification, and distribution with attribution and patent protection.
MIT License	Very permissive; allows reuse, modification, and distribution with minimal restrictions (just attribution).
BSD 2-Clause License	Permissive license similar to MIT; allows redistribution with attribution, no endorsement clause.
BSD 3-Clause License	Like BSD 2-Clause but adds a clause preventing use of contributors' names for promotion.
Boost Software License 1.0	Permissive license designed for C++ libraries; allows commercial and open use with attribution.
Creative Commons Zero (CC0) 1.0	Public-domain dedication; users can freely use, modify, and distribute without restrictions.
Eclipse Public License 2.0	Weak copyleft; modifications to EPL-covered code must remain open, but can link with proprietary code.

GNU General Public License v2.0 (GPL-2.0)	Strong copyleft; derivative works must also be licensed under GPL v2.
GNU General Public License v3.0 (GPL-3.0)	Strong copyleft with added protections against tivoization and patent issues.
GNU Lesser General Public License v2.1 (LGPL-2.1)	Weak copyleft; allows linking with proprietary software under certain conditions.
GNU Affero General Public License v3.0 (AGPL-3.0)	Like GPL v3, but requires source disclosure for network/server use.
Mozilla Public License 2.0 (MPL-2.0)	File-level copyleft; modified files must remain open, but larger projects can be proprietary.
The Unlicense	Places code in the public domain; no restrictions on use or distribution.

3. Licenses Comparison Table:

License	Commercial Use	Modify & Distribute	Copyleft Type	Attribution Required	Patent Protection	Source Disclosure Required
No License	No	No	None	No	No	No
MIT License	Yes	Yes	Permissive	Yes	No	No
Apache License 2.0	Yes	Yes	Permissive	Yes	Yes	No
BSD 2-Clause	Yes	Yes	Permissive	Yes	No	No
BSD 3-Clause	Yes	Yes	Permissive	Yes	No	No
Boost Software License 1.0	Yes	Yes	Permissive	Yes	No	No
Creative Commons Zero (CC0)	Yes	Yes	Public Domain	No	No	No
Eclipse Public License 2.0	Yes	Yes	Weak Copyleft	Yes	Yes	Yes (modified files only)
Mozilla Public License 2.0	Yes	Yes	Weak Copyleft	Yes	Yes	Yes (modified files only)
GNU LGPL v2.1	Yes	Yes	Weak Copyleft	Yes	No	Yes (library changes)
GNU GPL v2.0	Yes	Yes	Strong Copyleft	Yes	No	Yes

GNU GPL v3.0	Yes	Yes	Strong Copyleft	Yes	Yes	Yes
GNU AGPL v3.0	Yes	Yes	Strong Copyleft (Network)	Yes	Yes	Yes (including SaaS)
The Unlicense	Yes	Yes	Public Domain	No	No	No

4. List of GitHub Alternatives:

- GitLab Projects
- GitLab Groups
- GitLab Issues
- GitLab Merge Requests (MRs)
- GitLab CI/CD
- GitLab Issue Boards
- GitLab Wiki
- GitLab Releases
- GitLab Package Registry
- GitLab CI/CD Variables
- GitLab Pages
- GitLab Web IDE / Remote Dev
- GitLab Security Dashboard
- GitLab Dependency Scanning

5. GitHub vs GitLab Table:

Feature / Aspect	GitHub	GitLab
Primary Use	Code hosting and collaboration	Complete DevOps platform
Repository Hosting	Yes	Yes
Project Organization	Repositories & Organizations	Projects & Groups
Issue Tracking	GitHub Issues	GitLab Issues
Code Review	Pull Requests (PRs)	Merge Requests (MRs)
CI/CD	GitHub Actions (separate service)	Built-in GitLab CI/CD
Issue Boards	Limited (Projects/Boards)	Advanced Issue Boards
Wiki	Available per repository	Built-in Wiki
Releases	GitHub Releases	GitLab Releases

Package Registry	GitHub Packages	GitLab Package Registry
Pages / Hosting	GitHub Pages	GitLab Pages
Web IDE	GitHub Web Editor / Codespaces	Built-in Web IDE / Remote Dev
Security Features	Basic (Dependabot, security alerts)	Advanced Security Dashboard & Scanning
Dependency Scanning	Limited	Built-in Dependency Scanning

6. Open-Source vs Proprietary vs Freeware:

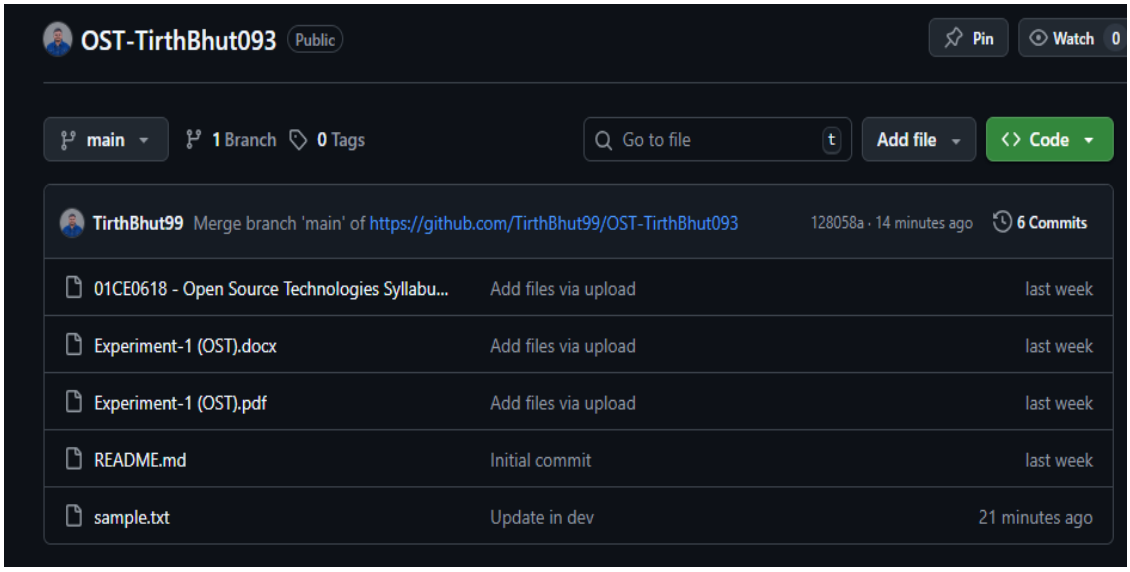
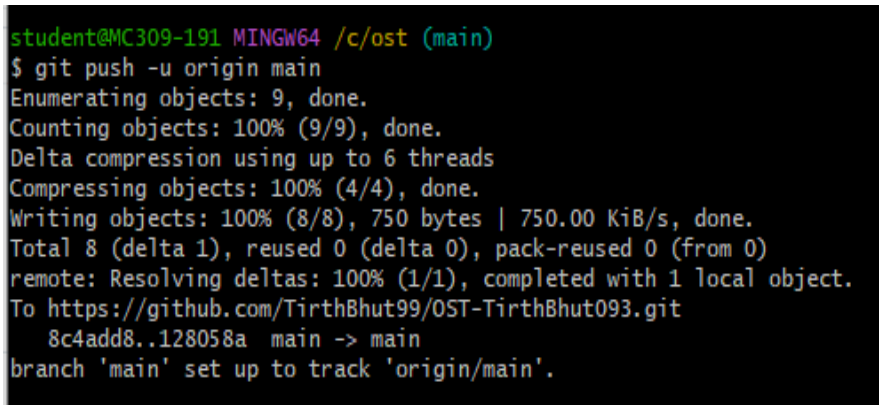
Aspect	Open-Source Software	Proprietary Software	Freeware
Source Code Access	Available to everyone	Completely closed	Closed
License Type	Open-source licenses (MIT, GPL, Apache)	Commercial/EULA	Vendor-specific
Cost of Software	Free or paid	Usually paid	Free
Upgrade Cost	Free updates	Paid upgrades	Free but limited
Customization Level	Fully customizable	Restricted	Not customizable
Vendor Lock-in	No vendor lock-in	High vendor lock-in	Medium
Security Auditing	Publicly auditable	Vendor-controlled	Vendor-controlled
Bug Fixing	Community-driven	Vendor-only	Vendor-only
Innovation Speed	Fast (community contributions)	Depends on company	Slow
Reliability	High (peer review)	High (official testing)	Moderate
Legal Freedom	High	Very low	Low
Commercial Use	Allowed	Allowed with license	Usually restricted

Experiment 2

AIM: Setup Git and explore commands related to Version Control System (VCS).

```
student@MC309-191 MINGW64 /c
$ git config --global user.name "Tirth Bhut"
$ git config --global user.email "tirthbhut07@gmail.com"

student@MC309-191 MINGW64 /c
$ git --version
git version 2.52.0.windows.1

student@MC309-191 MINGW64 /c
$ git init
Initialized empty Git repository in C:/.git/

student@MC309-191 MINGW64 /c
$ mkdir ost
student@MC309-191 MINGW64 /c
$ mkdir ost
mkdir: cannot create directory ‘ost’: File exists

student@MC309-191 MINGW64 /c
$ git init
Reinitialized existing Git repository in C:/.git/

student@MC309-191 MINGW64 /c
$ cd ost

student@MC309-191 MINGW64 /c/ost
$ git init
Initialized empty Git repository in C:/ost/.git/

student@MC309-191 MINGW64 /c/ost (master)
$ git status
On branch master
No commits yet
nothing to commit (create/copy files and use "git add" to track)

student@MC309-191 MINGW64 /c/ost (master)
$ echo sample.txt
sample.txt

student@MC309-191 MINGW64 /c/ost (master)
$ echo demo.py
demo.py

student@MC309-191 MINGW64 /c/ost (master)
```

```
$ git status
On branch master
No commits yet
nothing to commit (create/copy files and use "git add" to track)

student@MC309-191 MINGW64 /c/ost (master)
$ dir
student@MC309-191 MINGW64 /c/ost (master)
$ echo sample.txt
sample.txt

student@MC309-191 MINGW64 /c/ost (master)
$ dir

student@MC309-191 MINGW64 /c/ost (master)
$ echo Hello> sample.txt

student@MC309-191 MINGW64 /c/ost (master)
$ dir
sample.txt

student@MC309-191 MINGW64 /c/ost (master)
$ cat sample.txt
Hello

student@MC309-191 MINGW64 /c/ost (master)
$ git commit -m "Initial Commit"
On branch master
Initial commit
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        sample.txt
nothing added to commit but untracked files present (use "git add" to track)

student@MC309-191 MINGW64 /c/ost (master)
$ git config --global init.defaultBranch main

student@MC309-191 MINGW64 /c/ost (master)
$ git commit -m "Initial Commit"
On branch master
Initial commit
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        sample.txt
nothing added to commit but untracked files present (use "git add" to track)

student@MC309-191 MINGW64 /c/ost (master)
$ git add sample.txt
```

```
student@MC309-191 MINGW64 /c/ost (master)
$ git commit -m "Initial Commit"
[master (root-commit) 739b442] Initial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 sample.txt

student@MC309-191 MINGW64 /c/ost (master)
$ git branch
* master
student@MC309-191 MINGW64 /c/ost (master)
$ git branch -m master main

student@MC309-191 MINGW64 /c/ost (main)
$ git branch
* main

student@MC309-191 MINGW64 /c/ost (main)
$ git checkout main
Already on 'main'

student@MC309-191 MINGW64 /c/ost (main)
$ git checkout -b dev
Switched to a new branch 'dev'

student@MC309-191 MINGW64 /c/ost (dev)
$ echo Hello from dev>> sample.txt

student@MC309-191 MINGW64 /c/ost (dev)
$ git add.

student@MC309-191 MINGW64 /c/ost (dev)
$ git commit -m "Update in dev"
[dev 95837a9] Update in dev
 1 file changed, 1 insertion(+)

student@MC309-191 MINGW64 /c/ost (dev)
$ git checkout main
Switched to branch 'main'

student@MC309-191 MINGW64 /c/ost (main)
$ git merge dev
Updating 739b442..95837a9
Fast-forward
 sample.txt | 1 +
 1 file changed, 1 insertion(+)
student@MC309-191 MINGW64 /c/ost (main)
$ dir
sample.txt
```

```
student@MC309-191 MINGW64 /c/ost (main)
$ cat sample.txt
Hello
Hello from dev

student@MC309-191 MINGW64 /c/ost (main)
$ git remote add origin https://github.com/TirthBhut99/OST-TirthBhut093.git

student@MC309-191 MINGW64 /c/ost (main)
$ git remote -v

student@MC309-191 MINGW64 /c/ost (main)
$ git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 6 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (8/8), 750 bytes | 750.00 KiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/TirthBhut99/OST-TirthBhut093.git
   8c4add8..128058a  main -> main
branch 'main' set up to track 'origin/main'.
```